

A Tiered Compilation Architecture for Low-Latency Wasm Runtimes in Edge Computing

Master's Thesis Defense

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Motivation

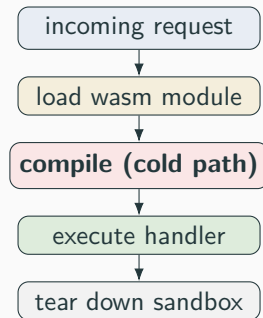
Edge computing reshapes the workload

Edge nodes face two tight budgets.

- **Response time:** tens of milliseconds per request.
- **Resources:** 1–2 GB RAM and few vCPUs.

The workload mix is short-lived, stateless functions that start and exit constantly.

WebAssembly fits the role. The format is compact, language-agnostic, sandboxed, and predictable. Fastly Compute, Cloudflare Workers, and Fermion Spin all run WebAssembly in production.



Per-request lifecycle.

The WasmEdge cold-start bottleneck

WasmEdge is the leading server-side Wasm runtime. Microsoft, ByteDance, and Sony all run WasmEdge in production.

WasmEdge compiles with LLVM end-to-end. The LLVM pipeline produces near-native code. The pipeline is also expensive:

- Compile time grows faster than module size.
- Memory usage during compilation is large.
- Every cold start pays the full compile cost.

The structural problem. A single-tier compiler treats every invocation the same way. A request that runs for one microsecond pays the same compile cost as one that runs for one minute.

A SpiderMonkey workload compiles for 90 seconds; the actual work takes 0.07 seconds. 4/39

Execution time is highly skewed

A small fraction of functions accounts for the bulk of run-time.

- Heavy optimization on cold functions wastes the compile budget.
- Cold functions receive optimization they do not need.
- Hot functions wait behind cold compilation before they can start.

The established remedy is tiered compilation.

- A **Tier-1** baseline compiles every function fast.
- A **Tier-2** optimizer recompiles hot functions later, guided by profile data.
- V8, JavaScriptCore, and SpiderMonkey all use this pattern.

WasmEdge has no baseline tier, no profile-driven promotion, and no mid-execution code replacement.

Browser-engine Wasm pipelines. V8 pairs Liftoff and TurboFan. JavaScriptCore runs IPInt, BBQ, and OMG. SpiderMonkey runs Baseline and Ion. All three use function-entry promotion. Public documentation reports no OSR between the baseline JIT and the optimizing JIT.

Server-side Wasm pipelines. Wasmtime offers Cranelift (optimizing) and Winch (baseline). The runtime does not currently tier up between them.

OSR in language VMs. HotSpot has supported OSR for over two decades. .NET 7 added OSR in 2022 with the one-shot-caller motivation this thesis also targets. Tracing JITs (PyPy, LuaJIT) sidestep OSR through trace recording.

This thesis fills a gap. No published server-side WebAssembly runtime combines a non-LLVM Sea-of-Nodes baseline, selective LLVM-frontend reuse, and loop-level OSR.

Problem Statement and Contributions

Three sub-problems

Goal. Cut cold-start latency without giving up the steady-state throughput that LLVM delivers on hot code.

Tier-1 baseline. Compile every wasm function fast at module load. Cover the full WebAssembly instruction set the workloads use. Reuse the existing WasmEdge runtime contracts.

Tier-2 optimizer. Reuse the LLVM frontend to recompile hot functions. Bridge the two tiers' calling conventions. Hide the LLVM compile cost from the user thread.

In-flight migration. A function called once spends its run-time inside one long loop. The prologue never re-fires, so entry-table promotion delivers no speedup. A separate mechanism must migrate the running frame mid-loop.

1. **Tier-1 baseline JIT for WasmEdge.**

Integrates a lightweight Sea-of-Nodes library. Uses one uniform calling convention for every function. Embeds per-function and per-loop counters directly in compiled code.

2. **Tier-2 selective LLVM-frontend recompilation.**

Synthesizes a mini-module per batch. Adds three ABI bridge thunks: `fwd_thunk`, `t1_thunk`, `entry_thunk`. Composes batches by walking up the call graph and admitting hot direct callees.

3. **On-Stack Replacement.**

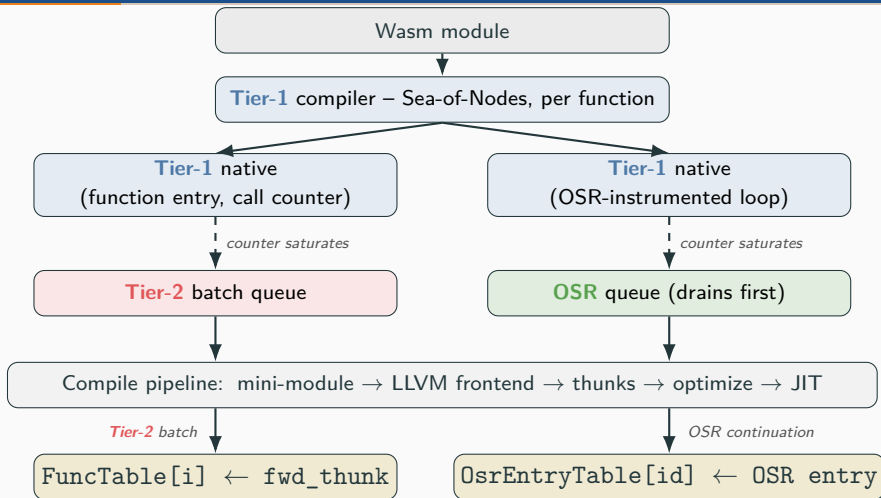
Instruments every outermost loop with a three-state back-edge diamond. Synthesizes a continuation that starts execution at the loop header.

4. **Empirical evaluation.**

Reports a step-by-step ablation on 39 strengthened sightglass kernels.

Architecture Overview

System architecture



Pipeline at a glance

- The user thread runs **Tier-1** native code and trips the embedded counters.
- A worker thread serves **Tier-2** batches and OSR continuations off the critical path.
- Two notification paths feed two queues: function-entry promotion and loop OSR.
- The worker drains the OSR queue first. An OSR continuation must publish before the running loop exits.
- The worker publishes each result with one atomic store into the dispatch slot.
- The pipeline coexists with the existing LLVM AOT path. Modules already compiled ahead of time skip **Tier-1**.

Tier-1: Baseline JIT

Tier-1: a lightweight Sea-of-Nodes baseline

Code generator: dstogov/ir. A small Sea-of-Nodes JIT library, descended from the Zend JIT in PHP 8.

- The library runs a fixed, linear pipeline. No pass manager.
- Compile latency stays bounded and predictable.
- The codebase is small enough to vendor and modify.

Three design choices distinguish the proposed tier.

1. Adopt the Sea-of-Nodes library as the code generator.
2. Translate wasm to IR in a single pass. The translator emits a PHI only for locals the loop actually modifies. Naive PHI-for-every-local inflates IR by an order of magnitude.
3. Embed profile counters directly inside compiled bodies. A saturated counter short-circuits in three IR operations.

Calling convention and JitExecEnv

Every **Tier-1** function takes the same two arguments.

$$f(\text{env}^*, \text{args}^*) \rightarrow \text{void}$$

The args buffer holds one 64-bit slot per wasm parameter. One dispatch shape covers wasm-defined functions, host imports, and indirect calls.

The JitExecEnv struct groups state into three sets.

Group	Contents
Lookup tables	FuncTable, linear memory, globals, indirect-call shadow
Helper addresses	host call, memory growth, profile-saturation helpers
Promotion state	call counters, back-edge counters, OsrEntryTable, OSR buffer

The worker thread writes only the OsrEntryTable pointer. The user thread owns every other field.

Tier-2: Selective LLVM Recompilation

The core idea: mini-module synthesis

Reuse the LLVM frontend instead of rewriting it. The frontend is mature and already lowers every wasm construct WasmEdge needs.

Mini-module synthesis (Algorithm 1)

1. Deep-copy the original wasm module.
2. Replace every defined function outside the batch with a small stub. The stub returns the function's declared type.
3. Run the LLVM frontend on the resulting mini-module.

Why default returns and not traps? A trap stub looks non-returning to LLVM. LLVM folds every call site into a trap and deletes the dispatch. The post-processing pass then has no call site left to rewrite. A default-return stub validates trivially and keeps every call site alive.

Three ABI bridge thunks

Tier-1 passes arguments in a flat 64-bit buffer. LLVM passes arguments in typed registers. Three thunks bridge the mismatch, one per direction of cross-tier dispatch.

Thunk	Direction	Role
fwd_thunk	Tier-1 → Tier-2	Lives in FuncTable after publication. Repacks Tier-1 arguments into typed parameters and calls the Tier-2 body.
t1_thunk	Tier-2 → Tier-1	Replaces a non-batch stub body inside the mini-module. Repacks typed arguments back into the Tier-1 buffer and dispatches under the Tier-1 convention.
entry_thunk	LLVM→ Tier-1	Sits in front of every Tier-1 function. Catches indirect calls from Tier-2 , OSR, or AOT code that target a Tier-1 function.

Each thunk handles one direction. Neither tier needs to know the other tier's ABI.

The compile-ordering invariant

Post-processing must run *before* the LLVM optimizer. Reversing the order produces wrong code.

Why the order matters. Before post-processing installs the bridges, every non-batch stub looks pure to LLVM. An aggressive pass inlines the stub's return constant into every call site and deletes the dispatch. Post-processing then finds no call site to rewrite.

Correct sequence.

1. Run the LLVM frontend unoptimized. Every call site survives.
2. Post-process: install `fwd_thunks`, rewrite stubs into `t1_thunks`, prune unused stubs.
3. Run the LLVM optimizer. The optimizer now sees real call targets.
4. Emit native code and publish the `fwd_thunk` pointer into `FuncTable`.

Batch composition: walk up, then bounded BFS

Promoting only the threshold-tripping function is not enough. The function that trips the counter is often a small leaf inside a larger hot caller. A leaf compiled alone still calls back into **Tier-1** on every helper, so the optimizer cannot inline across the boundary.

Algorithm 2 (sketch)

1. **Walk up one hop.** Pick the hottest direct caller as the batch root.
2. **Bounded BFS.** Traverse direct callees from the root up to a depth limit and a batch size limit.
3. **Two-test admission** for each candidate callee:
 - Dynamic test: the call counter exceeds zero.
 - Static test: the caller body references the callee at least twice.

The static test admits siblings whose dynamic counters lag behind but that the optimizer will still benefit from seeing.

On-Stack Replacement

The one-shot-caller problem

A function called once, spending its run-time inside one long loop.

This is the shape of cryptographic kernels, parsing loops, and many serverless handlers.

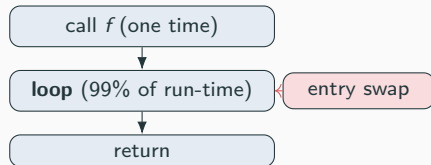
Function-entry promotion delivers no speedup.

- The running **Tier-1** frame finishes the loop and returns.
- The entry-table swap fires too late to matter.
- No future call ever observes the swap.

Measured at Step 2:

shootout-ctype $0.37\times$ LLVM,

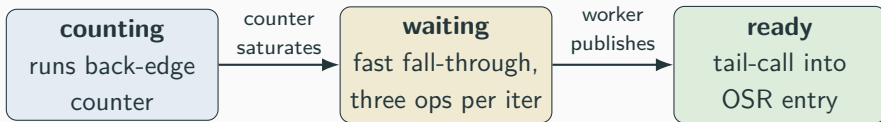
blind-sig $0.40\times$ LLVM.



swap arrives too late

Three-state back-edge instrumentation

Each outermost loop reads one slot of `OsEntryTable` on every back-edge. The slot encodes a three-state machine.



Steady-state cost. Three operations per iteration in the *waiting* state. An earlier two-array design needed six. Full pseudocode in the backup slides.

The continuation reuses the mini-module mechanism with three changes.

1. **Parameter list.** The new function takes the original parameters and the original locals, flattened into one list. The caller passes the running frame's state in through the call.
2. **Function body.** The body opens the enclosing control constructs, then resumes the original instruction stream from the loop header.
3. **Fresh function slot.** The continuation lives in a new slot. The original function stays untouched. Calls inside the continuation dispatch back to the original through `t1_thunk`.

Structural bail-out

Some loops cannot be safely re-entered mid-execution. The synthesis rejects those loops and the **Tier-1** path keeps running them correctly.

Why OSR continuations drain first

Two queues share one worker. The OSR queue has priority.

- A **Tier-2** batch redirects *future* calls. Late publication still helps.
- An OSR continuation migrates a *currently-running* frame.
The migration window closes when the running loop exits.
- An OSR continuation that lands after the loop returns is a no-op.

Function-entry promotion and OSR operate independently.

- The two mechanisms target different invocation shapes.
- One function may be entry-promoted to **Tier-2** *and* carry OSR continuations for its loops at the same time.

Evaluation

Benchmark suite: sightglass-strong.

Upstream sightglass kernels run in well under a second. The strengthened suite extends each kernel's **Tier-1**-only run-time into a 5–10 s band. **Tier-2** now has enough post-switch runway to amortize the worker's compile cost.

39 analyzed kernels group into five execution shapes.

- **Frequent-call** (16): one hot function called many times.
- **Hot-leaf** (6): the threshold-tripper sits below the real hot work.
- **One-shot-caller** (7): outer function called once, dominated by one loop.
- **Resistant** (4): too little run-time or no inline opportunity.
- **Compile-dominated** (7): LLVM compile cost dwarfs the wasm work.

Baselines. Whole-module LLVM JIT for the throughput ceiling; **Tier-1** alone for the latency floor.

Metrics. Work-time, compile time, end-to-end wall-clock.

Three configurations, three steps

Each step turns on one mechanism on top of the previous step.

Step	t1/t2	LLVM/t2	Mechanism added
Step 1 – promote hot callees only	0.97×	0.76×	function-entry promotion
Step 2 – walk-up + bounded BFS	1.03×	0.80×	better batch composition
Step 3 – on-stack replacement (full system)	1.11×	0.86×	in-flight migration

- Geometric mean across all 39 kernels.
- t1/t2: how much faster the full system runs than **Tier-1** alone.
- LLVM/t2: how close the full system gets to whole-module LLVM JIT.
A value of 0.86× means the full system recovers 86% of LLVM throughput.

Step 2: walk-up rescues hot-leaf kernels

Step 1 batched only the leaf that tripped the counter. The optimizer had nothing big enough to inline into. Step 2 rolls the surrounding caller into the batch.

Kernel	Step 1 LLVM/t2	Step 2 LLVM/t2	swing
hashset	0.62×	0.84×	+0.22
shootout-ed25519	0.38×	0.58×	+0.20
rust-json	0.62×	0.75×	+0.13
rust-compression	0.75×	0.87×	+0.12
regex	0.85×	0.95×	+0.10
pulldown-cmark	0.75×	0.84×	+0.09
gcc-loops	0.74×	0.83×	+0.09

Two groups still lag after Step 2. One-shot-caller kernels (shootout-ctype 0.37×, blind-sig 0.40×) need OSR. Compile-dominated kernels have flat steady-state; end-to-end wall-clock is the right metric for them.

Step 3: OSR closes the one-shot-caller gap

The seven kernels with the largest Step 2 → Step 3 jump all sit in the one-shot-caller group.

Kernel	Step 2 LLVM/t2	Step 3 LLVM/t2	swing
shootout-ctype	0.37×	0.88 ×	+0.51
blind-sig	0.40×	0.80 ×	+0.40
shootout-random	0.63×	1.00×	+0.37
shootout-base64	0.78×	0.99×	+0.21
shootout-matrix	0.75×	0.95×	+0.20
shootout-ratelimit	0.79×	0.96×	+0.17
shootout-minicsv	0.79×	0.95×	+0.16

What happens at the swing. The loop's back-edge counter saturates while the loop is still running. The worker compiles a continuation. The next iteration tail-calls into LLVM-optimized code.

Compile time and end-to-end wall-clock

Compile time. How long the runtime spends compiling before steady state.

End-to-end wall-clock. How long the user actually waits.

Compile-time gain

Geometric mean: **11.1**×

The full system finishes compilation about 11 times faster than whole-module LLVM JIT.

The **Tier-2** optimizer runs on the worker thread, off the user's critical path.

End-to-end gain

Geometric mean: **1.73**×

The full system finishes the end-to-end workload about 73% faster on average.

The gain comes from saved compile time, even though steady-state throughput is 14% lower.

The largest wall-clock wins

The wall-clock benefit is largest on compile-dominated workloads. These workloads are the real-world targets for the architecture.

Kernel	Tier-2 time	LLVM time	speedup
tinygo-json	473 ms	10 361 ms	21.9×
tract-onnx-image-classification	9.7 s	136 s	14.1×
spidermonkey-json	~9 s	~90 s	~10×
spidermonkey-markdown	~9 s	~90 s	~10×
spidermonkey-regex	~9 s	~90 s	~10×
image-classification	303 ms	2 667 ms	8.8×
sqlite3	1 985 ms	16 953 ms	8.5×

Why the wins cluster here. Each module embeds an interpreter, a reflection-driven dispatcher, or a graph of large tensor operators. The source-to-wasm toolchain has already baked in the hot path, so LLVM cannot recover further speedup. Saving the LLVM compile cost is therefore pure win.

Where the tiered design loses ground

Long-running synthetic kernels. The **Tier-1** throughput gap is no longer offset by a meaningful cold-start saving.

Kernel	end-to-end vs. LLVM
shootout-ctype	0.93×
shootout-sieve	0.84×
shootout-fib2	0.90×

The central observation. Workloads split bimodally. Compile-dominated workloads win big.

Long-running synthetic kernels lose a small amount. The architecture trades a modest steady-state cost for a large cold-start saving. The trade pays off most clearly on the real-world workloads the architecture targets.

Limitations and Future Work

No SIMD lowering. **Tier-1** does not lower wasm SIMD instructions. The translator rejects modules that use them. The **Tier-2** promotion gate accepts only scalar signatures.

No de-tiering. Promotion is one-way. A function promoted under one workload shape stays promoted even if the shape changes.

x86-64 only. The current prototype runs on x86-64. The library backend supports aarch64, but no aarch64 port has been validated.

Worker contention on small hosts. The evaluation used a 24-core workstation. On a single-vCPU edge node, the worker competes with user code and erodes the wall-clock gain.

SIMD lowering and wider promotion scope. Add SIMD lowering to **Tier-1**. Widen the OSR transition buffer for 128-bit slots. Relax the promotion gate to admit SIMD and reference-type signatures.

aarch64 port. Re-derive the bridge thunks for the AArch64 calling convention. Re-validate OSR continuation entry parameter layout against the aarch64 register allocator.

Refcounted JIT cache for de-tiering. Hold **Tier-1** code alive while a frame may still reach it. A function promoted under one workload shape could revert to **Tier-1** and re-promote later.

Profile-guided batch policy. Replace the fixed walk-up depth and BFS bound. Use call-site frequencies, branch biases, and value profiles to shape each batch.

Conclusion

Summary

This thesis presents a **tiered compilation architecture for WasmEdge** that separates cold-start latency from peak code quality.

Three design components.

- **Tier-1**: Sea-of-Nodes baseline JIT with embedded counters and a uniform calling convention.
- **Tier-2**: mini-module synthesis drives the LLVM frontend; three ABI bridge thunks; walk-up batch policy.
- **OSR**: three-state back-edge diamond; continuation resumes a running frame mid-loop.

Headline results across 39 kernels.

- Compile time: **11.1×** faster than whole-module LLVM JIT.
- Steady-state throughput: **86%** of LLVM JIT preserved.
- End-to-end wall-clock: **1.73×** faster on average; up to 21.9× on `tinygo-json`.
- OSR closes the one-shot-caller gap: `shootout-ctype` $0.37\times \rightarrow 0.88\times$.

Thank You

Questions?

Author: Hsiang-Yu Lee (李祥宇)

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Code: WasmEdge fork with tiered JIT

Backup: workload categorization (39 kernels)

Frequent-call (16) blake3-scalar, bz2, gcc-loops, hashset, pulldown-cmark, quicksort, regex, shootout-gimli, shootout-heapsort, shootout-keccak, shootout-memmove, shootout-seqhash, shootout-sieve, shootout-switch, shootout-xblabla20, shootout-xchacha20, tinygo-regex

Hot-leaf (6) blind-sig, rust-compression, rust-json, rust-protobuf, shootout-ctype, shootout-ed25519

One-shot-caller (7) blind-sig, shootout-base64, shootout-ctype, shootout-matrix, shootout-minicsv, shootout-random, shootout-ratelimit

Resistant (4) richards, rust-html-rewriter, shootout-fib2, shootout-nestedloop

Compile-dominated (6) image-classification, spidermonkey-json, spidermonkey-markdown, spidermonkey-regex, sqlite3, tinygo-json, tract-onnx-image-classification

shootout-ctype and blind-sig sit in two groups. Both mechanisms must fire to close the gap.

Backup: why default returns instead of trap stubs

The earlier design used trap stubs inside the mini-module. LLVM responded in three steps:

- LLVM marked each trap stub as non-returning.
- LLVM folded every call site that referenced a stub into a trap.
- LLVM deleted the dispatch path the post-processing pass needed to rewrite.

Default returns avoid the chain.

- Each stub returns the function's declared type. Validation succeeds.
- Post-processing rewrites the stubs into `__trap` before the optimizer runs. The call sites survive.

Backup: scope of JitExecEnv writes

Field ownership keeps cross-thread synchronization simple.

- The user thread writes every field except the `0srEntryTable` pointer.
- The worker thread writes only the `0srEntryTable` pointer.
- Publishing a **Tier-2** batch is one atomic store into `FuncTable[i]`.
- Publishing an OSR continuation is one atomic store into `0srEntryTable[id]`.

No locks sit on the hot dispatch path. On the supported host architecture, the dispatch load already provides the ordering the design needs.

Backup: full pseudocode of the back-edge diamond

```
slot = LOAD OsrEntryTable[loopId]
if slot != WAITING:                // sentinel 0
    if slot == COUNTING:           // sentinel 1
        counter = LOAD BackEdgeCounters[loopId]
        if counter < threshold:    // unsigned!
            counter += 1
            STORE BackEdgeCounters[loopId] = counter
            if counter == threshold: notify_osr(loopId)
    else:                          // READY: slot holds an entry pointer
        snapshot wasm locals into per-thread buffer
        return slot(env, locals)    // tail-call the OSR entry
fall through to the back-edge
```

The unsigned comparison matters. A signed comparison would treat the saturated value as negative and let the gate pass. The counter would wrap and re-fire notification on every threshold-th invocation.

Backup: threats to validity

- **Single-host measurement.** All numbers from one 24-core workstation. Behavior on production edge hosts is not measured.
- **Three-run median.** Not a tail-latency or p99 analysis.
- **No CPU contention scenario.** The worker thread runs on a dedicated core in the experiments. Single-vCPU contention is acknowledged in the limitations but not measured.
- **Sightglass-strong, not cross-runtime.** The suite compares successive configurations of one runtime. No cross-runtime claim is made.
- **x86-64 only.** The aarch64 port is not validated.

Backup: OSR lineage

HotSpot pattern. A per-method back-edge counter triggers an OSR compile. The OSR compile emits a method version whose entry point is the loop header. Live state transfer uses a stack map. HotSpot has supported OSR for over two decades.

.NET 7 (2022). Added OSR with the same one-shot-caller motivation this thesis targets. The release notes describe methods called once that spend their run-time inside loops.

This thesis. Follows the HotSpot and .NET-7 lineage. Three simplifications:

- Restricted to outermost-loop back-edges.
- Live state is the wasm-level local set, not a stack map.
- Continuation is a fresh wasm function compiled through the LLVM frontend.

WebAssembly's structured control flow makes the simplifications safe.

Backup: why dstogov/ir, not Cranelift

Three reasons dstogov/ir fits the baseline tier role.

- **Bounded, predictable compile latency.** The library runs a fixed, linear pipeline with no pass manager. Compile cost scales linearly with function size.
- **SSA-quality output from a single-pass frontend.** The IR is SSA by construction. The thesis does not need a separate SSA-construction phase.
- **Small enough to vendor and modify.** The runtime can extend the library when a backend change is needed.

Why not Cranelift. Cranelift's pipeline targets higher code quality at higher compile cost. Cranelift is closer to an optimizing tier than a baseline tier. The codebase is also larger and harder to vendor.

Backup: bibliography highlights

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